COLLEGE OF ENGINEERING
KEVIN T. CROFTON DEPARTMENT OF AEROSPACE AND OCEAN ENGINEERING
VIRGINIA TECH

PROJECT HESTIA

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COSMIC THE POWER OF COLLABORATION

VT LUNA

Project Description

Our Project Aims To:

- Demonstrate creating landing pad tiles for future Artemis missions
- Demonstrate usage of lunar regolith with additive manufacturing through sintering

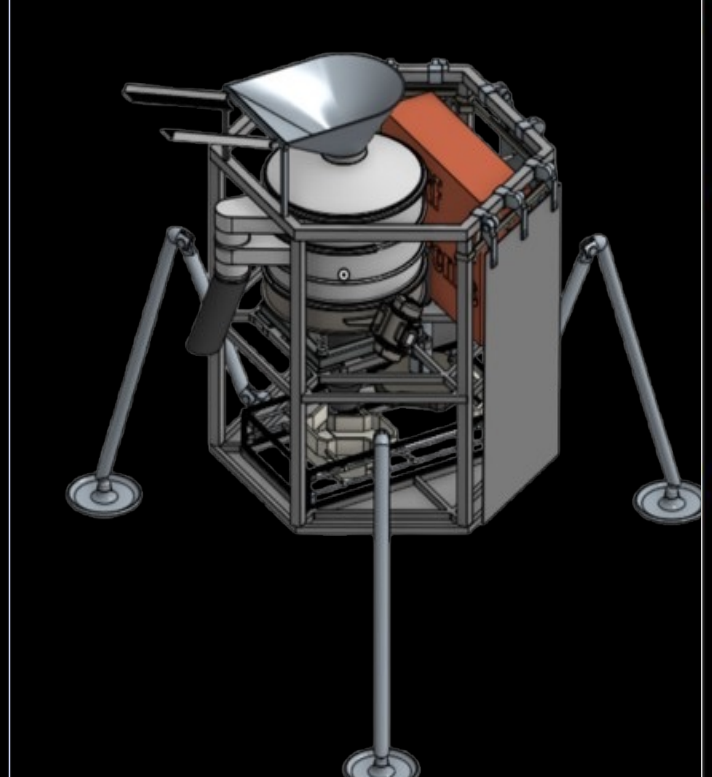
Rover

- Regolith Collection
- Deposits into STiM
- Collection Rate: 1kg/min
- Bucket Volume: 5.65 L

Stationary Tile Maker (STiM)

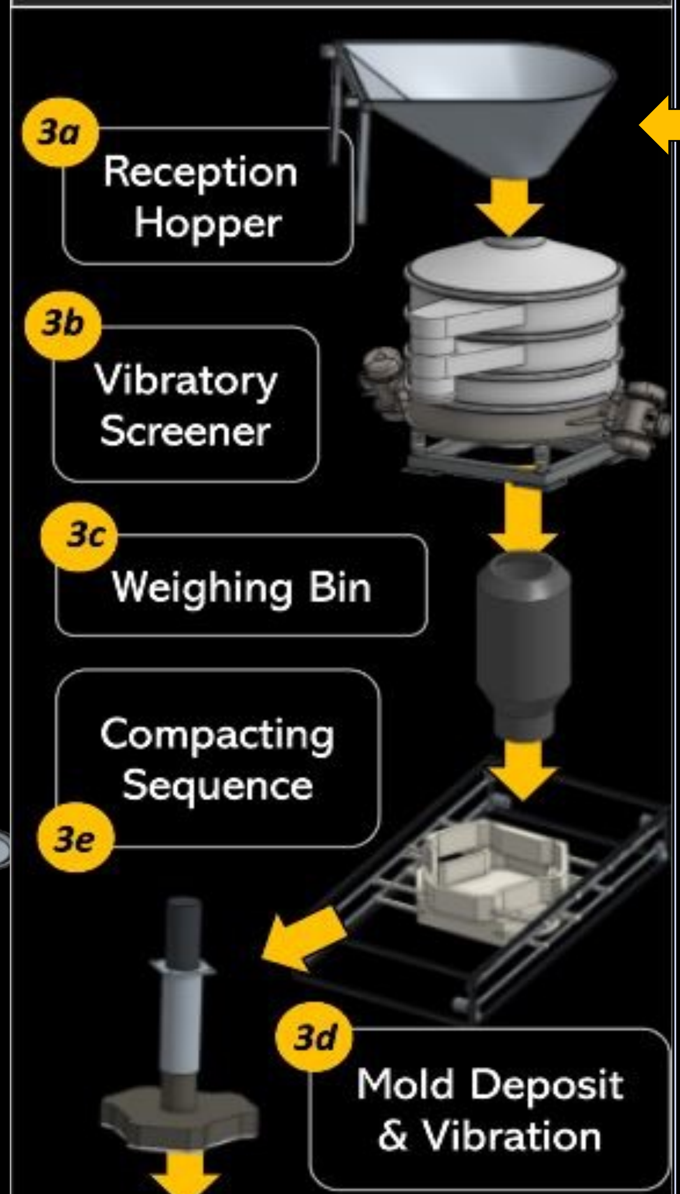
STiM:

- Filter Regolith
- Vibrate and Compress to Mold
- Microwave Sinter in Layers to Form Tiles

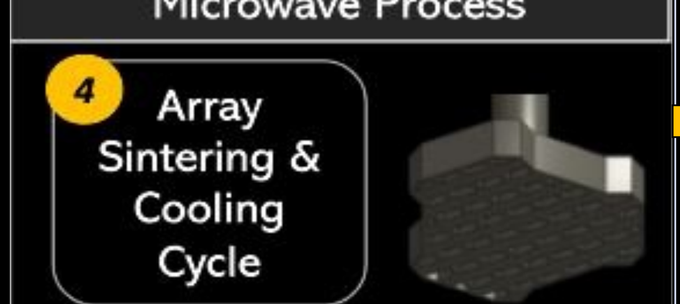


TILE FORMATION

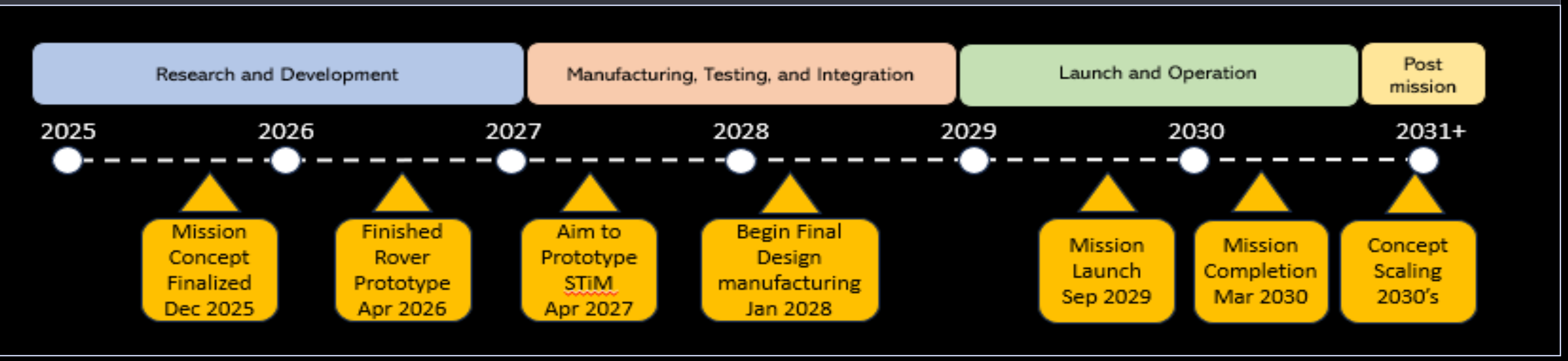
Regolith Conditioning (Flow of Regolith)



Microwave Process



Mission Timeline



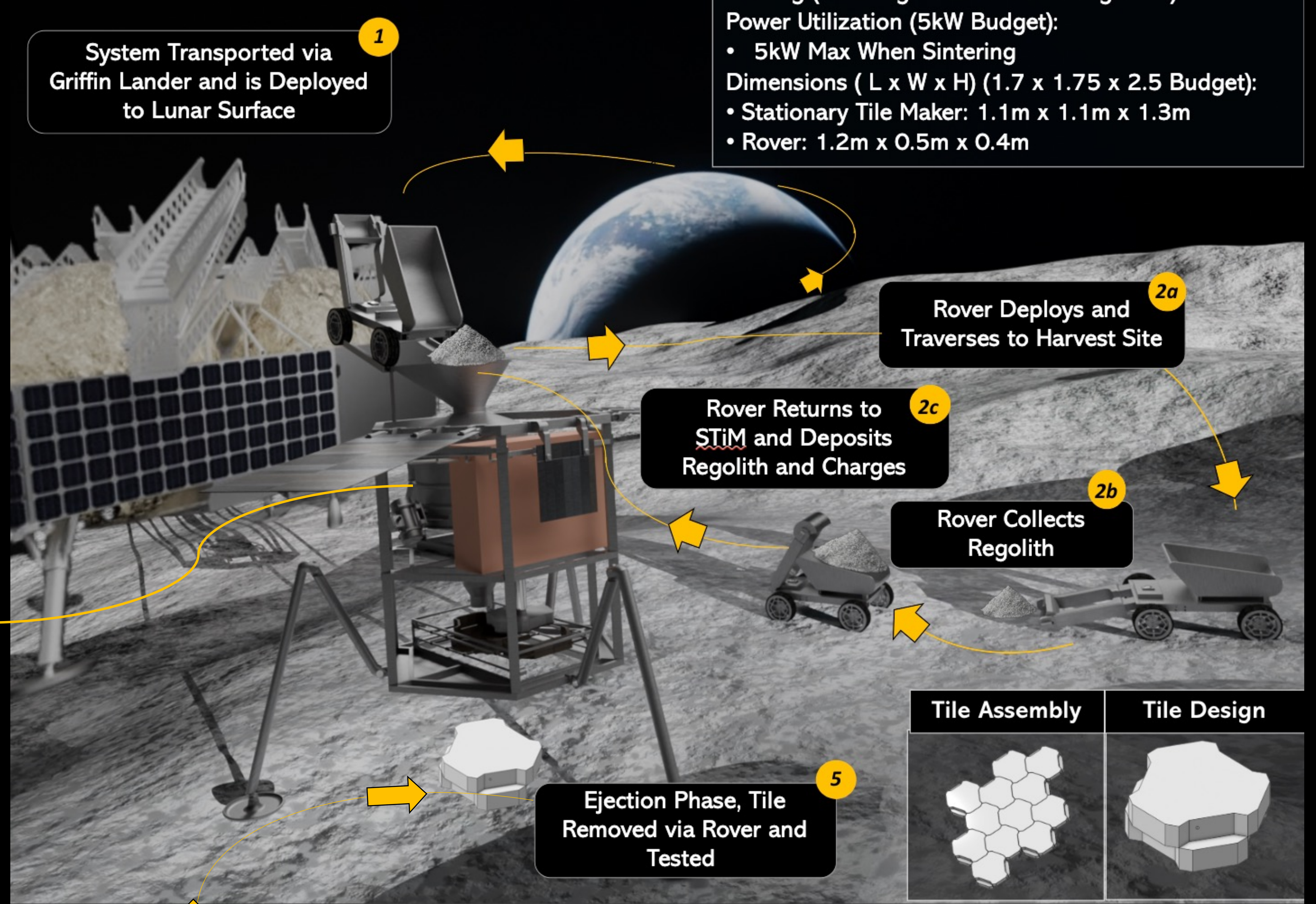
CONCEPT OF OPERATIONS

Key Specifications:

Payload Mass (Mass Budget of 200 kg):
- 184kg (29.24kg Rover + 154.96kg STIM)

Power Utilization (5kW Budget):
- 5kW Max When Sintering

Dimensions (L x W x H) (1.7 x 1.75 x 2.5 Budget):
- Stationary Tile Maker: 1.1m x 1.1m x 1.3m
- Rover: 1.2m x 0.5m x 0.4m



Microwave Sintering

Why Microwave Sintering:

- In-situ manufacturing
- Scalable, reuseable, and controllable
- Low mass requirement
- 915 MHz wave guide antenna
- 50 VDC
- 63% efficiency

Tile Design

- Designed to resist heating and compression
- Interlock with low precision manipulation
- Production Time: 7.2 hrs/tile
- 0.3m radius, 0.106m thick
- ~450 tiles during mission timeline
- Launch Pad Area: 25 sq meters

Assumptions

Griffin Lander used for:

- Payload Delivery
- Power
- Communications

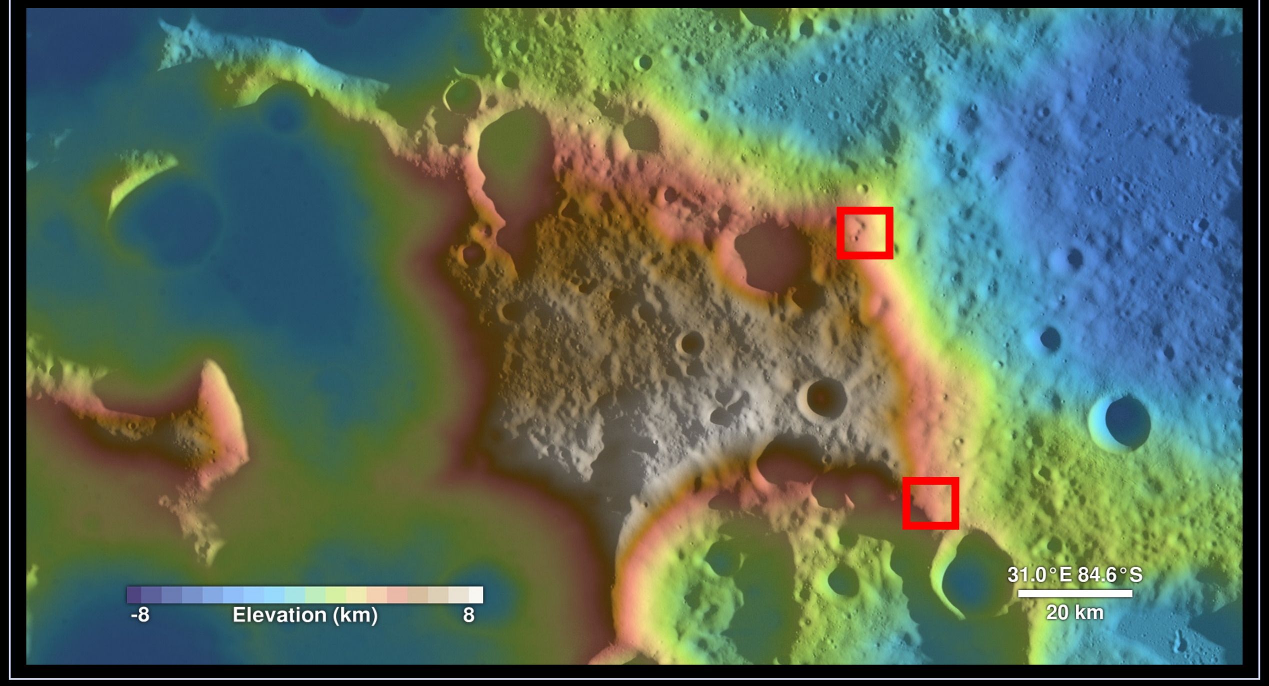
Astronauts used to:

- Retrieve and test tiles on Earth

Landing Site

Mons Mouton Plateau was Chosen for:

- Flat, Navigable Terrain
- Long Durations of Continuous Illumination
- Ideal Regolith Composition



Cost and Mass Budget